

6.11 Thermal characteristics

The operating junction temperature T_J must never exceed the maximum given in [Table 24: General operating conditions](#).

The maximum junction temperature in °C that the device can reach if respecting the operating conditions, is:

$$T_J(\text{max}) = T_A(\text{max}) + P_D(\text{max}) \times \Theta_{JA}$$

where:

- $T_A(\text{max})$ is the maximum operating ambient temperature in °C,
- Θ_{JA} is the package junction-to-ambient thermal resistance, in °C/W,
- $P_D = P_{\text{INT}} + P_{\text{I/O}}$.
 - P_{INT} is power dissipation contribution from product of I_{DD} and V_{DD}
 - $P_{\text{I/O}}$ is power dissipation contribution from output ports where
 $P_{\text{I/O}} = \sum (V_{\text{OL}} \times I_{\text{OL}}) + \sum ((V_{\text{DDIO1}} - V_{\text{OH}}) \times I_{\text{OH}})$, taking into account the actual V_{OL} / I_{OL} and V_{OH} / I_{OH} of the I/Os at low and high level in the application.

Table 78. Thermal resistance

Symbol	Parameter	Package ⁽¹⁾	Value	Unit
Θ_{JA}	Thermal resistance junction-ambient	TSSOP20	81	°C/W
		WLCSP24	92.2	
		UFQFPN28	57.2	
		UFQFPN32	42	
		LQFP32	51.4	
		UFQFPN48	31.9	
		LQFP48	51.7	
		LQFP64	43	
		UFBGA64	66.5	
Θ_{JB}	Thermal resistance junction-board	TSSOP20	54.7	°C/W
		WLCSP24	60.7	
		UFQFPN28	22.2	
		UFQFPN32	23.9	
		LQFP32	28.5	
		UFQFPN48	16.2	
		LQFP48	28.8	
		LQFP64	25.3	
		UFBGA64	49.3	